

A large language model-based approach for fake review detection: the implicit characteristics perspective

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ABSTRACT

Detecting fake online reviews is crucial for maintaining the integrity and sustainable development of digital platforms. Prior work has focused on overt linguistic signals, but the implicit traits of deceptive content remain largely untapped, thus constraining detection accuracy. This study proposes a novel fake review detector called LLMIC, which leverages large language models (LLMs) and introduces complex system paradigms to quantify the implicit characteristics. Specifically, we propose a new multifractal method XSMF to explore multi-faceted fractal patterns of reviews, and a new recurrence plot approach MSFRP to analyze multi-layered recurrence behaviors of reviews. Extensive experiments demonstrate that LLMIC outperforms dozens of recent models, and also confirm the effectiveness of XSMF and MSFRP. In addition, we provide new insights: fake reviews tend to exhibit larger Hurst exponents, broader Hölder exponents, and more prominent recurrence diagonals, reflecting their greater complexity and variability.

1. Introduction

Online reviews offer insights that transcend mere product descriptions and marketing content, and have become a pivotal force in shaping consumer trust and reputation of brands (Hajek et al., 2023; Mohawesh et al., 2024). Positive reviews can significantly enhance sales volumes and brand image (Salminen et al., 2022; Hajek et al., 2023), while negative reviews drive consumers to seek alternatives, thereby prompting businesses to improve their products or services. A recent survey in 2024 reveals that over 90% of consumers affirm that reviews significantly shape their overall perception of brands, and 58% of consumers express a willingness to pay a premium for products associated with highly-rated brands¹. Therefore, the credibility and authenticity of reviews have become crucial, as the presence of fake or manipulated reviews will severely undermine the healthy development of online platforms.

A vast amount of fake review detection research has emerged, primarily favoring two paradigms, feature-engineering approaches, and language model-based approaches. Feature engineering methods focus on extracting explicit features, which are typically direct and observable measurements, such as sentiment & semantic score (Cao et al., 2022; Hajek et al., 2023), topic & POS tags (Birim et al., 2022), and behavior patterns (Narayan et al., 2022). Language model-based approaches focus on the representations learned via deep learning techniques (Duma et al., 2024), such as text-image neural representations, sentence representations, etc., (Neisari et al., 2021; Shunxiang et al., 2023; Bikku et al., 2024; Rahim et al., 2024); or they fine-tune pre-trained language models, such as

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¹ <https://www.businessdasher.com/online-review-statistics/>

Transformer-RoBERTa (Mohawesh et al., 2024), knowledge-enhanced BERT with TextCNN (Lu et al., 2023), and complexity-augmented DistilBERT (Zhang et al., 2024a), etc., (Han et al., 2023; Salminen et al., 2022; Liu et al., 2023; Mohawesh et al., 2023; Hyder et al., 2024; Fan et al., 2025).

Despite these advancements, detectors that overlook the implicit characteristics of reviews may constrain the detection capabilities. Fake reviews are often designed to deceive users by superficial mimicry, e.g., focusing on similar POS distributions, sentences with comparable representations². While this strategy can be effective for simple cases, it becomes increasingly challenging as the complexity of fake reviews grows. The potential for differentiation lies in the implicit characteristics, which are more difficult to replicate. This arises from the fundamentally divergent underlying intentions: fake reviews are generated for deceptive or illicit purposes, whereas real reviews are rooted in authentic user experiences. Here, the implicit characteristics refer to the intrinsic dynamics of an object itself, typically at the macro-level. These attributes are neither isolated nor static (Kwapien & Drożdż, 2012; Cimini et al., 2019). For example, local self-similarities and singularities quantified by Hölder exponent series reflect the internal temporal correlation structures of an object (Wang et al., 2024b). By shifting attention to the implicit characteristics, we can unlock discriminative power that existing methods cannot provide.

This study proposes to analyze and quantify the implicit characteristics of reviews for fake review detection. A promising and widely accepted paradigm for this is the framework of complex systems. By modeling a review itself as a complex system, we aim to analyze its concealed, macro-level patterns through the dynamic behavior of its micro-level components (e.g., tokens), thereby uncovering implicit characteristics (Stanisz et al., 2024). This is inspired by insights from Gödel, Escher, Bach: *An Eternal Golden Braid*, which illuminates how complex phenomena arise from the recursive and hierarchical organization inherent in systems, seeking to decipher the principles whereby simple rules can spawn sophisticated behaviors.

In complex systems research, fractals, often eulogized as the "fingerprints of God"³, are among the most widely used concepts, which depict self-similarity and long-range dependencies within systems, such as those found in text (De-Santis et al., 2024; Stanisz et al., 2024). Typically, text is firstly transformed into embeddings using models like BERT or Word2Vec, and then is averaged to produce a one-dimensional numerical sequence. This sequence undergoes further analysis through detrended scaling, culminating in the quantification of the text's fractal dimensions (Wang et al., 2023b; Wang et al., 2024b). Another favored concept is the recurrence plot (RP) (De-Santis et al., 2024; Stanisz et al., 2024), which graphically represents the recurrence of states within a system relative to previous states, thus revealing underlying periodicities or aperiodic patterns within data that might initially appear stochastic or disordered. Existing methodologies generate RPs by transforming text embeddings into one-dimensional numerical sequences, subsequently assessing the relational distances between points (Purohit & Panigrahi, 2024). These methods have notable limitations. First, translating reviews into one-dimensional sequences risks omitting details captured in the embeddings. Second, the sensitivity of RPs to individual data points introduces a level of uncertainty when abstracting reviews into numerical form.

To address these challenges, two methods are crafted to model the implicit characteristics of reviews, including a novel multifractal method XSMF (Spaces of Multi-Fractal, X representing the various spaces involved), and a novel Multi-series Scale Fuzzy Recurrence Plot (MSFRP) method. Specifically, XSMF treats review embeddings as multidimensional sequences, designing dynamic scaling across multiple directions to derive novel fractal-based spaces; MSFRP employs fuzzy computing to assess embeddings across multiple series, and analyzes local motifs with different scales, thereby yielding multiple recurrence-based patterns.

Building upon these, this study proposes an LLM-based approach for fake review detection using Implicit Characteristics (LLMIC). Firstly, LLM is leveraged to expand reviews and generate the numerical sequences. The benefits are two-folds. LLM generates reviews with human-like insights; and the synthetic reviews augment existing datasets to enhance detection models. Then, these sequences are characterized by XSMF and MSFRP to capture the implicit characteristics, as discussed previously. Finally, these characteristics are fused into neural networks to predict whether a review is fake or real.

Extensive experiments demonstrate the state-of-the-art performance of LLMIC. We also examine the effectiveness of XSMF and MSFRP, and the implicit characteristics. The sensitivity of XSMF and MSFRP is also analyzed. Moreover, we investigate the differences between fake and real reviews in terms of implicit characteristics, uncovering several novel findings. For instance, fake reviews exhibit a larger Hurst exponent, indicating a greater degree of persistence and self-similarity; the Hölder exponent of fake reviews is broader, reflecting increased variability in their local behaviors; the recurrence diagonal lines in fake reviews are more pronounced, suggesting stronger repetitive patterns.

The main contributions are as follows.

1. This study proposes to introduce implicit characteristics into fake review detection, and designs an LLM-based approach, which achieves state-of-the-art performance.
2. Two methods XSMF and MSFRP, are proposed to quantify multi-faceted and multi-level fractal patterns and recurrence behaviors in reviews.
3. A series of new insights is provided on the difference between fake reviews and real ones in terms of implicit characteristics, e.g., larger Hurst exponents, broader Hölder exponents, and more pronounced recurrence diagonals in fake reviews.

² [https://www.wsj.com/tech/personal-tech/google-search-shopping-results-fake-reviews-7795b2b9?](https://www.wsj.com/tech/personal-tech/google-search-shopping-results-fake-reviews-7795b2b9?mssockid=247cd839a3766f130a4ccde9a2616ebb)

³ <https://www.pbs.org/wgbh/nova/physics/hunting-hidden-dimension.html>

2. Related work

The field of fake review detection has garnered increasing attention. Existing methods include two primary approaches: language model-based methods and feature engineering-based methods, as summarized in Table 1.

Language model-based methods typically involve the fine-tuning and enhancement of (pre-trained) language models. Some studies argue that language models (such as LSTM) can capture language dependencies within reviews, which is crucial for distinguishing real from fake content, as the dependencies often encode certain cues about authenticity (Han et al., 2023). For instance, Bathla et al., (2022) introduce an aspect-replica learning mechanism, integrating CNN and LSTM; and Liu et al. (2022) propose a hierarchical attention model that combines N-grams, CNNs, and Bi-LSTMs. Other works shift the primary detection stage to pre-trained language models by fine-tuning them on augmented datasets generated via popular text generation models. For example, Salminen et al. (2022) and Liu et al. (2023) employ GPT-2 to synthesize training set and subsequently fine-tune RoBERTa. Beyond these methods, some researchers find that the performance of pre-trained models can be further enhanced through the integration of domain-specific or task-oriented language models. For instance, Lu et al. (2023) combine TextCNN with SKEP (Sentiment Knowledge Enhanced Pre-training) to enrich BERT representations; and Mohawesh et al. (2024) incorporate Transformer and LSTM layers following RoBERTa embeddings to capture temporal and structure dynamics.

Feature engineering-based approaches seek to identify features that can differentiate fake reviews from real ones. Some studies have demonstrated that linguistic features often serve as reliable indicators, as fake reviews tend to exhibit distinct textual patterns compared to genuine content. For instance, fake reviews frequently contain a higher proportion of adjectives and adverbs and tend to display more uniform word counts (Budhi, Chiong, Wang, & Dhakal, 2021). So, some methods apply machine learning algorithms to uncover various linguistic features (Cao et al., 2022; Singhal & Kashef, 2023). Another line of research posits that sentiment within reviews can reveal important authenticity cues. Genuine consumers, who are emotionally invested in their reviews that often exhibit more context-sensitive sentiment expressions, compared to individuals generating fake reviews for profit or manipulation (Hajek et al., 2020; Birim et al., 2022). For instance, Hajek et al. (2023) measure sentiment scores associated with different products and their aspects in Amazon reviews; Shunxiang et al. (2023) apply positive-unlabeled learning to train neural networks based on the sentiment intensity distributions across fake and real reviews. In addition to linguistic and sentiment features, several studies investigate

Table 1
Survey on fake review detection studies.

Studies	Datasets	Main ideas	Detection models
Language model-based			
Bathla et al., 2022	Hotel, Yelp	To combine CNN and LSTM on aspect-replica learning	CLSTM-ASP
Liu et al., 2022	Hotel, Restaurant, Doctor	Hierarchical attention of N-gram, CNN and Bi-LSTM	DSRHA
Han et al., 2023	Hotel	Knowledge-enhanced CNN and LSTM	EKI-SM
Salminen et al., 2022	Amazon, Hotel	Data augmentation based on GPT-2 and ULMFit	OpenAI model, FakeRoBERTa, NBSVM
Liu et al., 2023	Amazon	Data augmentation based on GPT-2 and OPT	AugRoBERTa, AugSVMFT
Lu et al., 2023	Hotel, Restaurant, Doctor	To integrate BERT, TextCNN, and SKEP	BSTC
Mohawesh et al., 2023	Yelp, Hotel	To combine SLP, MLP and CNN based on ELECTRA, ELMo and GPT2	EEG-SMC
Hyder et al., 2024	Hotel, Yelp	To integrate layer combination strategy into BERT	DeceptiveBERT
Mohawesh et al., 2024	Hotel, Restaurant, Doctor	To combine Transformer and LSTM on RoBERTa	TransLSTM-RoBERTa
Zhang et al., 2024b	Amazon	To fuse DistilBERT, BiLSTM, and CNN dynamically	DynDistilBERT-BiLSTM-CNN
Feature engineering-based			
Budhi et al., 2021	Yelp	Linguistic and behavioral features	SVM, MLP
Cao et al., 2022	Hotel, Restaurant, Doctor	Linguistic and semantic analysis of local context and temporal patterns	ST-MFLC
Singhal & Kashef, 2023	Hotel, Deception	Linguistic features based on weighted stacking ensemble modeling	WSEM
Hajek et al., 2020	Hotel, Restaurant, Doctor, Amazon	Multi-granularity sentiment analysis	DFFNN, DFSCNN
Birim et al., 2022	Amazon	Sentiment, topic, cluster and purchase analysis	RFTD-VP, DT-VP, NN-VP, SVM-VP
Hajek et al., 2023	Amazon	Sentiment analysis of product aspects and categories	LDA-ABSA
Shunxiang et al., 2023	Hotel, Yelp	Sentiment intensity on positive-unlabeled learning	SIPUL
Narayan et al., 2022	Amazon	Graph-based user-text interactions	GraphSAGE-Co
Xu et al., 2022	Hotel	Graph-based consistency features across content, sentiment and time	WSUN
Sharifpour & Lahmiri, 2023	Amazon	Behavior, sentiment, and their discrepancy features	ISIGRU
Duma et al., 2024	Yelp	Interactive information, and heterogeneous graph convolutional network	DFIFM
Rahim et al., 2024	Amazon	Aspect-based sentiment analysis, and synthetic minority over-sampling	IABSA-SS
Neisari et al., 2021	Hotel, Restaurant, Doctor	Self-organizing map-based text-image representations	SOM-CNN
Bikku et al., 2024	Amazon	Quantum neural network representation	QuanNN

behavioral patterns as distinguishing features, such as employing graph-based computations to extract user-text interactions, and review ratings (Narayan et al., 2022; Xu et al., 2022; Sharifpour & Lahmiri, 2023). Notably, a separate line of research is inspired by interdisciplinary techniques, which aims to capture potentially more effective features. For example, Neisari et al. (2021) utilize a self-organizing map to convert text into representations, matching each word with specific pixels in an image. The resulting patterns serve as features of review authenticity. Also, Bikku et al. (2024) explore the application of quantum neural networks for feature extraction.

Despite the diversity of these models, gaps remain. LLMs with their transformative potential and wide-ranging impact across industries, have yet to be explored for fake review detection. Furthermore, the implicit characteristics of reviews, such as those that can be examined through the lens of complex systems, remain uncharted territory.

3. Method

This study introduces a new fake review detection approach LLMIC, which is a deep learning architecture that integrates the strengths of LLMs, the multifractal perspective of XSMF and the recurrence view of MSFRP, as illustrated in Fig. 1. The role of LLMs is twofold: (1) Data augmentation of reviews: augmenting the training data by generating review variants; (2) Embedding generation of reviews: transforming reviews into high-dimensional embeddings, which serve both as inputs to LLMIC's deep learning backbone and as signals for XSMF and MSFRP. The obtained results are connected and jointly optimized end-to-end. This dual use of the embeddings establishes a direct operational link between LLMs and the implicit-characteristic modules. For a better understanding, we first elucidate XSMF, then explain MSFRP, and finally provide an overview of LLMIC framework, where the parameter symbols across subsections are not shared.

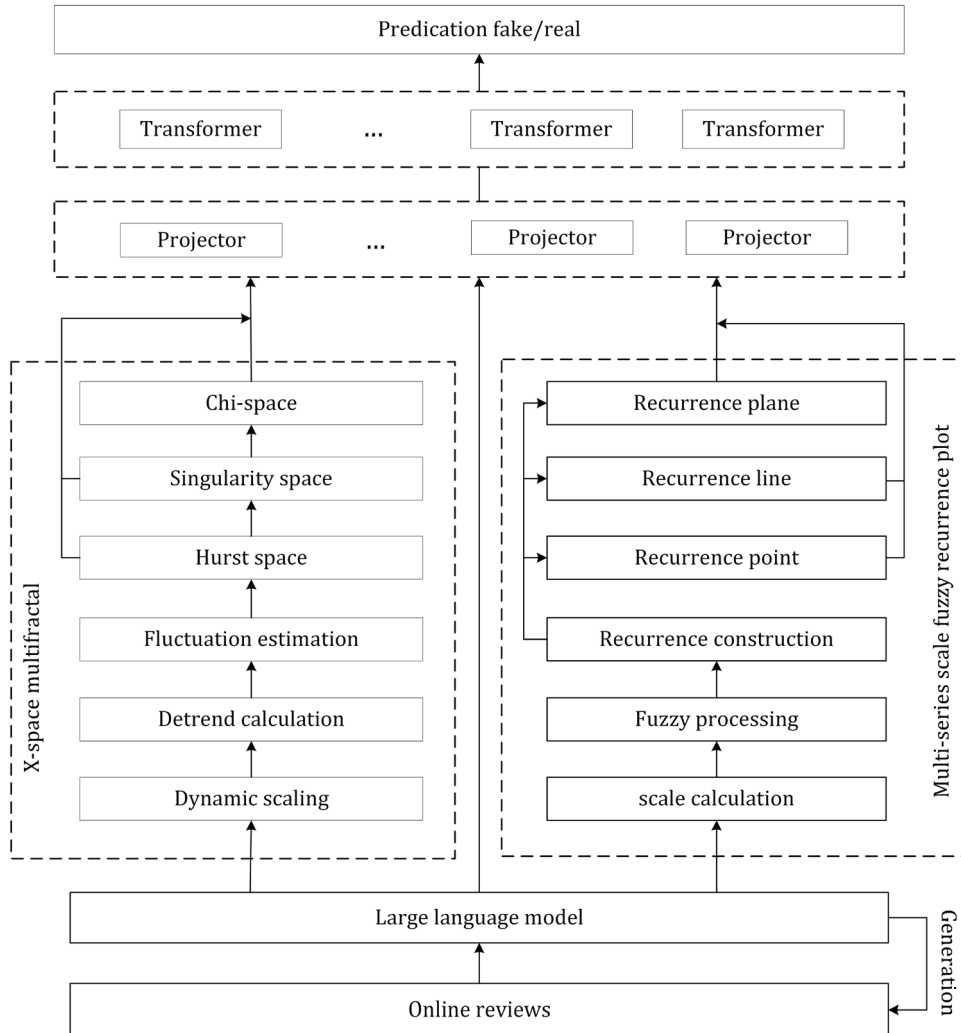


Fig. 1. Architecture of LLMIC.

3.1. X-space multifractal

The proposed XSMF method takes a review's embeddings as input and outputs multiple fractal-related spaces, referred to as 'x'. Let Y represent the embedding matrix where $Y \in \mathbb{R}^{N \times D}$, with N representing the number of tokens and D representing the dimensions: $Y = [e_{ij}], i = 1, \dots, N; j = 1, \dots, D$. The transformation process begins by converting Y into the profile matrix E using $E(i, j) = \sum_{i=1}^i \sum_{j=1}^j (Y(i, j) - \langle Y \rangle)$. We then partition E into submatrices $E_{m,l}$ of scale $\delta = \{\delta_N \times \delta_D\}$, i.e., $E = U_{m=1}^M U_{l=1}^L E_{m,l}; M = \frac{N}{\delta_N}, L = \frac{D}{\delta_D}$.

Considering the significance of δ , from an embedding feature perspective, smaller scales might reveal insights about local regularities; larger scales provide a global view of scaling behaviors. From a fractal perspective, smaller scales can detect unique details in multifractality; larger scales, encompassing more data points, tend to offer statistically more stable estimates of fractal indices. Therefore, we dynamically select δ , aiming to emphasize scales with higher energy. Specifically, for the token or dimension direction series $x(t)$ with index t and frequency f , we first compute its Fourier transform $X(f) = FFT(x(t), t, f)$. We then calculate the normalized power spectral density $PSD(f) = \text{norm}(X(f) \cdot \bar{X}(f)), f > f'$, with the evolutionary complex conjugate $\bar{X}(f)$ (Hong, 2023), and filter out non-positive frequencies, using an empirical threshold $f' = 1e-13$ to focus on physically meaningful PSD values. Subsequently, we apply a threshold κ (empirically set at $1e-3$) to select frequencies, and, ensure the uniqueness and sort the resulting scales: $\delta =$

$$\text{sort} \left(\text{unique} \left(\left\{ \delta_i \mid \delta_i = \frac{1}{f_{\text{selected},i}}, f_{\text{selected}} = \{f \mid PSD(f) \kappa\} \right\} \right) \right).$$

After obtaining δ , we construct the matrix $U_{m,l}(i, j) = \sum_{k=1}^i \sum_{h=1}^j E_{m,l}(k, h)$, $(i, j) \in [1, \delta_N] \times [1, \delta_D]$ for each segment by integrating over tokens and dimensions. Hence, we execute $R_{m,l}(i, j) = U_{m,l}(i, j) - \text{argmin}_P \|U_{m,l} - P\|_2^2(i, j)$ to remove potentially unrelated trends, where P represents the space of bivariate polynomials.

For each detrended segment $R_{m,l}$, we evaluate its fluctuation measures from different aspects: Standard-deviation-based, $F_{m,l}^{sd}$ (Eq. 1), Skewness-based, $F_{m,l}^{sk}$ (Eq. 2), Kurtosis-based, $F_{m,l}^{ku}$ (Eq. 3):

$$F_{m,l}^{sd} = \left(\frac{1}{\delta_N \delta_D} \sum_{i=1}^{\delta_N} \sum_{j=1}^{\delta_D} [R_{m,l}(i, j) - \mu_{m,l}]^2 \right)^{1/2}, \mu_{m,l} = \frac{1}{\delta_N \delta_D} \sum_{i=1}^{\delta_N} \sum_{j=1}^{\delta_D} R_{m,l}(i, j) \quad (1)$$

$$F_{m,l}^{sk} = \left| \frac{\frac{1}{\delta_N \delta_D} \sum_{i=1}^{\delta_N} \sum_{j=1}^{\delta_D} [R_{m,l}(i, j) - \mu_{m,l}]^3}{(F_{m,l}^{sd})^3} \right| \quad (2)$$

$$F_{m,l}^{ku} = \frac{\frac{1}{\delta_N \delta_D} \sum_{i=1}^{\delta_N} \sum_{j=1}^{\delta_D} [R_{m,l}(i, j) - \mu_{m,l}]^4}{(F_{m,l}^{sd})^4} \quad (3)$$

We formulate the generalized fluctuation function (Eq. 4) across scales and order q :

$$F_q(\delta_N, \delta_D) = \left(\frac{1}{M \times L} \sum_{m=1}^M \sum_{l=1}^L [F_{m,l}^{sd} + F_{m,l}^{sk} + F_{m,l}^{ku}]^q \right)^{1/q} \quad (4)$$

As the order q (typically, $q \in [-10, 10]$) changes, the power law is emerged by $F_q(\delta_N, \delta_D) \sim (\delta_N \delta_D)^{H(q)}$. Thus, we use Boltzmann machine to process F_q , thereby determining the generalized Hurst exponents $H(q)$, where $\{q, H(q)\}$ forms the Hurst space. On this basis, we leverage the Legendre–Gauss quadrature rule to ascertain $H(q)$ (Rahimkhani et al., 2024), thereby obtaining the Hölder exponent series $\alpha = d[qH(q) - D_f]/dq$, where $D_f = 2$ is the topology dimension for E . So, we can obtain the singularity spectrum $f(\alpha) = q\alpha - qH(q)|_{\alpha=q(q)}$, and $\{\alpha, f(\alpha)\}$ can be as the singularity space.

Next, we can deduce the Rényi's generalized dimension Eq. 5, and normalize it through logistic transformation $\Psi(\cdot)$. Further, we have Eq. 6, where, $\text{flatten}(\cdot)$ is numerical flatten function, $\text{sort}(\cdot)$ is the ascending sorting function, and the $\Phi^{-1}(\cdot)$ is inverse cumulative distribution function of the standard normal distribution.

$$A_{LD} = \Psi \left(\frac{1}{q} \lim_{\delta_N, \delta_D \rightarrow 0} \frac{\log \sum_{m=1}^M \sum_{l=1}^L [F_{m,l}(\delta_N, \delta_D)]^q}{\log(\delta_N \delta_D)} \right) \quad (5)$$

$$D_G = \sqrt{\sum_{i=1}^{N \times D} \left(\text{sort}(\Psi(\text{flatten}(E)))_i - \Phi^{-1} \left(\frac{i - 0.5}{N \times D} \right) \right)^2}, \quad (6)$$

We conceptualize (A_{LD}, D_G) as the Chi-space, a novel perspective in the review multifractal. Diverging from previous methods that primarily utilize the width of the multifractal spectrum (WMS) to complement D_G (Mangalam & Kely-Stephen, 2024), our A_{LD} can capture review's Rényi distribution insights, which could offer a fresh impetus. Another consideration is that the α has to some extent

covered the fractal information carried by WMS.

Ultimately, XSMF can calculate the Hurst space, Singularity space, and Chi-space for each review. The pseudocode for XSMF is outlined in Algorithm 1.

Algorithm 2.

3.2. Multi-series scale fuzzy recurrence plot

The proposed MSFRP method takes review's embeddings as inputs and outputs quantifications of recurrence behaviors.

We treat the embeddings as multi-series across dimensions: $V^j = \{V_i^j \in \mathbb{R}^d | i = 1, 2, \dots, n_j\}$ for $j = 1, 2, \dots$, and to capture local and dynamic trends, we perform the scaling operation. Specifically, we first get a set of scaling sizes $S = \{s_1, s_2, \dots, s_K\}$ such as $\{2, 3, \dots\}$, where s_k represents the window size at scale k , and K is the total number of scales. Then we set the number of windows $N_{j,k} = \lfloor (n_j - s_k) / \Delta \rfloor + 1$, where Δ is the window shift, such as 5, and $\lfloor \cdot \rfloor$ denotes the floor function. We finally obtain the scaled sub-series, see Eq. 7:

$$V_s^j = \bigcup_{k=1}^K \bigcup_{m=1}^{N_{j,k}} \left\{ V_{(i)}^j | i = (m-1)\Delta + 1, \dots, (m-1)\Delta + s_k \right\} \quad (7)$$

To consider the potential uncertainty of sub-series, we perform fuzzy computing to enhance the robustness of recurrence analysis (Korać et al., 2025), see Eq. 8. Each sub-series V_k^j is processed to construct a fuzzy set $\lambda_k^j = (\phi_k^j, \psi_k^j)$ with $\psi_k^j = \frac{1}{k} \sum_{i=1}^k V_k^j(i)$ representing the mean value, ϕ_k^j defining the boundaries (the variance value) of the fuzzy set.

$$\lambda_k^j(x) = \begin{cases} 1 - \frac{|x - \psi_k^j|}{\phi_k^j} & \text{if } |x - \psi_k^j| < \phi_k^j \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

Next, we construct the recurrence condition, and define the fuzzy recurrence between two points i and j , see Eq. 9:

$$FRP_{i,j}^k = \Theta \left(\varepsilon - \left\| \sum_{m=1}^M (V_k^j(i)_m - V_k^j(j)_m) \right\| \times (1 - \lambda_k^j(i) \times \lambda_k^j(j)) \right) \quad (9)$$

Here, $\Theta(\cdot)$ is the Heaviside step function, with the threshold value ε that: when it is not lesser than the Euclidean norm term, a recurrence instance is formed, visualized as a point in a plot, otherwise marked as non-recurrence instance that is an empty. This approach scales Euclidean distance by the product of the fuzzy membership between V_k^j , making the recurrence condition more or less

Algorithm 1

XSMF.

Input: embedding matrix Y of review

Parameter: κ, f, Q
Output: x-space = {Hurst, Singularity, Chi}

- 1 PSD = $\emptyset$
- 2 for each x in rows(Y) $\cup$ columns(Y):
- 3 PSD(f) = norm(FFT(x))
- 4 PSD $\cup$ (f for f where PSD(f)[$f > f$] $> \kappa$)
- 5 δ = sorted_unique(round($1/|f|$)) for f in PSD
- 6 $E[i, j] = \sum_{u \leq i, v \leq j} (Y[u, v] - \text{mean}(Y))$
- 7 $F_{sd}, F_{sk}, F_{ku} = \emptyset, \emptyset, \emptyset$
- 8 for δ_N, δ_D in δ :
- 9 for m in range(0, N, δ_N):
- 10 for l in range(0, D, δ_D):
- 11 W = window($E, m, \delta_N, l, \delta_D$)
- 12 R = detrend(matrix(W), order = poly2)
- 13 sd = sqrt(mean(($R - \text{mean}(R)$)²)); $F_{sd} \cup sd$
- 14 sk = abs(mean(($R - \text{mean}(R)$)³) / sd^3); $F_{sk} \cup sk$
- 15 ku = mean(($R - \text{mean}(R)$)⁴) / sd^4 ; $F_{ku} \cup ku$
- 16 for q in Q :
- 17 $F_q = (\text{mean}((F_{sd} + F_{sk} + F_{ku})^q))^{1/q}$
- 18 $H(q)$ = Boltzmann(log $\Sigma(F_q)$, log F_q)
- 19 $\alpha(q)$ = $d/dq [q \cdot H(q) - 2]$
- 20 $f(\alpha)$ = $q \cdot \alpha - q \cdot H(q)$
- 21 Hurst_space = zip($q, H(q)$)
- 22 Singularity_space = zip($\alpha, f(\alpha)$)
- 23 A_{LD} = logistic(lim $_{\delta \rightarrow 0} \log \Sigma(F_q) / \log \delta^q$)
- 24 D_G = sqrt(Σ_i (sort(logistic(E)) _{i} - $\Phi^{-1}((i - 0.5) / (N \cdot D))$)²)
- 25 Chi_space = zip(A_{LD}, D_G)
- 26 return {Hurst: Hurst_space, Singularity: Singularity_space, Chi: Chi_space}

Algorithm 2

MSFRP.

Input: embedding series V of review

Parameter: S, Δ, η
Output: RPoint, RLine, RPlane

```

1 subseries =  $\emptyset$ 
2 for  $v$  in  $V$ :
3   for  $s$  in  $S$ :
4     subseries  $\cup$  (window( $v, s, \Delta$ ))
5   for  $w$  in subseries:
6      $\mu = \text{mean}(w)$ ;  $\sigma = \text{var}(w)$ 
7    $M = \text{len}(\text{subseries})$ 
8    $\text{dist} = \text{zeros}(M, M)$ 
9   for  $i$  in range( $M - 1$ ):
10    for  $j$  in range( $M - 1$ ):
11       $\text{euc} = \text{norm}(\text{subseries}[i] - \text{subseries}[j])$ 
12       $\lambda = (1 - |\text{subseries}[i] - \mu_i| / \sigma_i) \cdot (1 - |\text{subseries}[j] - \mu_j| / \sigma_j)$ 
13       $\lambda = \text{clip}(\lambda, 0, 1)$ 
14       $\text{dist}[i, j] = \text{euc} \cdot (1 - \lambda)$ 
15       $R = (\text{dist} \leq \eta \cdot \text{std}(\text{dist})).\text{astype}(\text{int})$ 
16      RPoint = mean( $R$ )
17       $P(l) = \text{histogram}(R, \text{direction}='diag')$ 
18       $P(v) = \text{histogram}(R, \text{direction}='vert')$ 
19      DET =  $\sum l \cdot P(l)$  for  $l \geq l_{\min} / \sum l \cdot P(l)$  for all  $l$ 
20      ENT =  $-\sum \text{norm}(P(l)) \cdot \log \text{norm}(P(l))$ 
21      LAM =  $\sum v \cdot P(v)$  for  $v \geq v_{\min} / \sum v \cdot P(v)$  for all  $v$ 
22      TT =  $\sum v \cdot P(v)$  for  $v \geq 2 / \sum P(v)$  for  $v \geq 2$ 
23      RLine = [DET, ENT, LAM, TT]
24      RPlane = CNN.ReLU( $R$ )
25 return {RPoint: RPoint, RLine: RLine, RPlane: RPlane}
```

stringent based on the degree of membership. Given the integration of RP into a deep learning training, heuristic methods are deemed inappropriate. Thus, ε is set as a multiple η of the standard deviation of Euclidean distances, for example, $\eta = 2$.

We calculate Recurrence point and Recurrence line, and propose Recurrence plane, across MSFRP layers (De-Santis et al., 2024; Stanisiz et al., 2024).

Recurrence point (RPoint = $\frac{1}{N^2} \sum_{i,j=1}^N FRP_{i,j}$, where N is the number of observed values) quantifies the density of recurrences, serving as an indicator of the chaotic behavior of the reviews. It reflects the underlying order within the apparent randomness of review's dynamical properties.

Recurrence line (RLine) consists of two types. The first is the diagonal lines, which indicate the embedding index periods where the trajectory revisits a previous state, suggesting periodicity or cyclic behavior in the dynamical system. We have Determinism ($DET = \frac{\sum_{l=l_{\min}}^N l \cdot P(l)}{\sum_{l=1}^N l \cdot P(l)}$) and Entropy ($ENT = \sum_{l=l_{\min}}^N P(l)_{\text{norm}} \cdot \log(P(l)_{\text{norm}})$), where $P(l)_{\text{norm}} = P(l) / \sum_{l=l_{\min}}^N P(l)$, $P(l)$ represents the histogram of diagonal line lengths. The second is the vertical/horizontal lines, which reflect periods during which the state of the system remains relatively unchanged, indicating trapping or laminar states. We have Laminarity ($LAM = \frac{\sum_{v=v_{\min}}^N v \cdot P(v)}{\sum_{v=1}^N v \cdot P(v)}$), Trapping Time ($TT = \frac{\sum_{v=v_{\min}}^N v \cdot P(v)}{\sum_{v=1}^N P(v)}$), where $P(v)$ is the histogram of vertical/horizontal line lengths. Hence, RLine = [DET; ENT; LAM; TT].

Recurrence plane (RPlane = $f(\sum_{m,n} W_{m,n} \cdot FRP_{i+m, j+n} + b)$) reflects the overall layout of review recurrence behavior, where we utilize a convolutional neural network to scan the recurrence plot, and $f(\cdot)$ is the activation function (e.g., ReLU), W is the trainable weight of the (m, n) -th matrix (filter), b is the trainable bias term.

The MSFRP's pseudocode is outlined as follows.

3.3. LLMIC

The LLMIC model is structured into three stages.

Stage 1: We prompt LLM to generate an extended set (X'_{real} and X'_{fake}) based on the training set (X_{real} and X_{fake}), see Eq. 10.

$$X'_{\text{real}} = \text{LLM}(X_{\text{real}}, \text{prompt } 1); X'_{\text{fake}} = \text{LLM}(X_{\text{fake}}, \text{prompt } 2) \quad (10)$$

For generating real reviews, we prompt the LLM with each real review from the training set to create additional real-like reviews. Note that truly authentic reviews are intrinsically elusive, as they can only arise from genuine user or consumer experiences with the service or product. As detailed in Prompt Method 1, we ensure that X'_{real} align closely with X_{real} in terms of conveyed information, such as content, style, form, and presentation. To preserve authenticity, we prompt the LLM to inhabit the mindset of an actual user, letting

it channel genuine feedback as much as possible, rather than fabricate it. This strategy has been shown to significantly enhance the LLM's ability to immerse itself in specific task contexts, as role-playing leverages the model's training on vast and diverse datasets to replicate scenarios effectively (Wang et al., 2025a). We also emphasize ethical considerations, ensuring that the generated reviews do not contain misleading information.

For generation of fake reviews X'_{fake} , we guide the LLM using X_{fake} as input, as described in Prompt method 2. Here, X'_{fake} is designed to closely match X_{fake} in terms of style, form, and presentation etc., while focusing on the subtle nuances of fake reviews rather than overtly fraudulent indicators. To provide a more contextual and more diverse training dataset for detection models, the content of X'_{fake} is allowed to deviate from X_{fake} . To minimize content repetition between fake and real reviews, X'_{fake} is not generated based on X_{real} . We balance the dataset by generating equal numbers of real and fake reviews. The size of X' is kept consistent with X , to prevent model overfitting to the idiosyncrasies of generated reviews.

Prompt method 1: Real review generation

You are a real user and consumer. Based on your understanding, please create a new real review that closely follows the original real review in terms of content and presentation:

1. Sentiment Consistency: The overall and local sentiments should match the original review.
2. Writing Style: Preserve the same style (e.g., formal, casual, enthusiastic, critical and so on).
3. Structure and Length: Maintain a similar structure and word count, including sentence complexity and paragraph organization and so on.
4. Linguistic Features: Replicate the use of adjectives, adverbs, and specific vocabulary etc., that reflect the original review's language.
5. Coherence and Relevance: Ensure the review is coherent, logically structured, and directly relevant to the product or service being reviewed.
6. Specificity and Detail: Include the same specific details or personal experiences mentioned in the original review to enhance authenticity.
7. Avoid Redundancy: Do not reuse exact phrases or sentences from the original review to ensure the generated review is not a direct copy.

Content Consistency:

- Do not introduce new features, benefits, or criticisms that were not present in the original review.
- Ensure that all details remain consistent with the original review, avoiding any alterations that could change the review's meaning or authenticity.

Ethical Considerations:

- Ensure that the generated review does not contain misleading information.
- The review should be plausible and reflective of genuine user feedback without fabricating false claims.

Original Real Review:

—

Please generate one new real review below that adheres to the above guidelines:

Prompt method 2: Fake review generation

Based on your understanding, please generate a new fake review given an original fake review. The new review should maintain consistency in the following aspects:

1. Sentiment: The overall and local sentiments should match the original review.
2. Style: Maintain the same writing style (e.g., formal, casual, enthusiastic, critical and so on).
3. Structure: Preserve the structural elements such as sentence length, paragraph organization, and use of headings or bullet points if present.
4. Specific Linguistic Features: Replicate any unique stylistic features and linguistic features such as vocabulary complexity, use of adjectives/adverbs, sentence complexity, and any unique phrases or idioms.
5. Coherence and Relevance: Ensure the review is coherent and remains relevant to the product or service being reviewed.

Content Variation:

- While maintaining the above surface characteristics, the actual content (specific details, experiences, examples) should differ.
- Incorporate different aspects of the product/service that were not mentioned in the original review.
- Avoid repeating exact phrases or sentences from the original review to ensure diversity.

Additional Guidelines:

- Avoid Overt Indicators of Fakeness: Do not include exaggerated praise, irrelevant information, or obvious grammatical errors. The review should appear as though it were written by a genuine user.
- Balanced Feedback: If the original review includes both positive and negative feedback, maintain a similar balance in the new review.

- Contextual Consistency: Ensure that any specific details or examples provided are plausible and contextually consistent with the product or service.

Original Fake Review:

—

Please generate one new fake review below that adheres to the above guidelines:

Stage 2: We first utilize LLM to compute the embeddings E_i of reviews $X = X + X' = \{x_i\}_{i=1}^N$, which can carry rich feature

information. We then calculate the implicit characteristics (IC) of reviews based on E_i , including Hurst space (IC_{Hurst}), Singularity space ($IC_{Singularity}$), Chi-space (IC_{Chi}) for XSMF's fractal quantification; and recurrence point (IC_{RPoint}), recurrence line (IC_{RLine}), and recurrence plane (IC_{RPlane}) for MSFRP's recurrence quantification.

$$IC_{Hurst,i} = \{(q_k, H(q_k))\}_{k=1}^Q; IC_{Singularity,i} = \{(\alpha_j, f(\alpha_j))\}_{j=1}^A; IC_{Chi,i} = \{(A_{LD,l}, D_{G,l})\}_{l=1}^L$$

$$IC_{MSFRP,i} = [IC_{RPoint,i}, IC_{RLine,i}, IC_{RPlane,i}]$$

To jointly process all implicit characteristics (each regarded as a modality) and capture the interdependencies, we develop a multi-modal Transformer. For each modality z , we define projection functions $v_i^{(z)} = \phi^{(z)}(f_i^{(z)})$ with a non-linear activation such as ReLU (Wang et al., 2022a), where $f_i^{(z)}$ is the representation of modality z . Specifically, for the Hurst space representation, Singularity space representation, and Chi-space representation, we reshape them into their respective sequences of vectors (H_i, S_i, C_i), and have:

$$H_i = \begin{bmatrix} [q_1, H(q_1)]^\top \\ \vdots \\ [q_Q, H(q_Q)]^\top \end{bmatrix}, v_i^{(h)} = \phi^{(h)}(H_i) = \text{Linear}^{(h)}(H_i)$$

$$S_i = \begin{bmatrix} [\alpha_1, f(\alpha_1)]^\top \\ \vdots \\ [\alpha_A, f(\alpha_A)]^\top \end{bmatrix}, v_i^{(s)} = \phi^{(s)}(S_i) = \text{Linear}^{(s)}(S_i)$$

$$C_i = \begin{bmatrix} [A_{LD,1}, D_{G,1}]^\top \\ \vdots \\ [A_{LD,L}, D_{G,L}]^\top \end{bmatrix}, v_i^{(c)} = \phi^{(c)}(C_i) = \text{Linear}^{(c)}(C_i)$$

For the $IC_{MSFRP,i}$, we treat it as a single vector $v_i^{(r)} = \phi^{(r)}(IC_{MSFRP,i}) = \text{Linear}^{(r)}(IC_{MSFRP,i})$.

Furthermore, considering that the embeddings E_i of reviews inherently encompass a wealth of feature information, we derive $v_i^{(e)} = \phi^{(e)}(E_i) = \text{Linear}^{(e)}(E_i)$.

Hence, we construct a combined sequence $V_i = [v_i^{(CLS)}; v_i^{(h)}; v_i^{(s)}; v_i^{(c)}; v_i^{(r)}; v_i^{(e)}]$ from all modalities, where $v_i^{(CLS)}$ is a commonly-used learnable [CLS] token for classification. Given the existence of different modalities, we add segment sequences $S_i = [s_{CLS}; s_h^{(1:Q)}; s_s^{(1:A)}; s_c^{(1:L)}; s_r; s_e^{(1:T_i)}]$ to distinguish them, where T_i represents the number of tokens in the i -th review. Also, as there is inherent sequentiality within each modality, we incorporate positional encodings (randomly initialized for different positions) $P_i = [p_1; p_2; \dots; p_{T_i^{total}}]$ for the sequence positions, where T_i^{total} is the sequence length of V_i .

We process $\tilde{V}_i = V_i + S_i + P_i$ using L Transformer layers (e.g., $L = 24$) to capture interdependencies, $H_i^{(l)} = \text{Transformer}^l(H_i^{(l-1)})$, $H_i^{(0)} = \tilde{V}_i$, $l = 1, 2, \dots, L$, which has been shown to be effective (Xu et al., 2024), where, each layer includes multi-head self-attention, facilitating the synthesis of diverse informational streams. Finally, we obtain a unified representation $R_i = \text{Norm}(H_i^{(L)})$ through layer normalization.

Stage 3: Considering that fake review detection is a binary classification task, we use the common Cross-Entropy classification loss that ensures the model learns to classify reviews correctly. Specifically, for the i -th review, we apply a linear layer to map the representation to category logits, $z_i = W_c R_i + b_c \in \mathbb{R}^K$, $K = 2$ (number of categories: real or fake); then, we can get the distribution, $\hat{y}_{i,k} = \text{softmax}(z_i)$, which represents the predicted probability that review i belongs to category k ; finally, we have Eq. 11, where $y_{i,k}$ is the ground truth label ($y_i \in \{0, 1\}$ represents the real and fake), N is the total number of reviews.

$$L_{CE} = - \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log(\hat{y}_{i,k}) \quad (11)$$

In addition, we construct a contrastive learning framework to better highlight the differences between fake and real reviews. The goal is to bring similar reviews closer in the representation space while pushing dissimilar reviews apart. For each anchor review representation $R(An_i)$, we randomly select a review from the same category, that is, the positive sample representation $R(Po_i)$, $y(An_i) = y(Po_i)$, and randomly select a review from the other category, that is, the negative sample representation $R(Ne_i)$, $y(An_i) \neq y(Ne_i)$. We encourage the distance between the anchor and positive representation to be smaller than the distance between the anchor and negative representation by a margin ϑ , as formulated in Eq. 12, where, $\|\cdot\|$ denotes the Euclidean distance, $[x]_+ = \max(0, x)$.

$$L_{CL} = \frac{1}{N} \sum_{i=1}^N [\|R(An_i) - R(Po_i)\|^2 - \|R(An_i) - R(Ne_i)\|^2 + \vartheta]_+ \quad (12)$$

The total loss combines the two losses, i.e., $L_{Total} = L_{CE} + \lambda L_{CL}$, where λ is a trainable parameter. L_{Total} is minimized to update the model parameters.

Table 2

Statistical information of datasets, where the vocabulary size is the total number of unique words.

Dataset	Word count		Vocabulary size		Sentence count		Review count	
	fake	real	fake	real	fake	real	fake	real
Hotel	117309	120731	6355	7214	7466	9332	800	800
Amazon	622550	830458	20471	28081	60004	73901	10500	10500

4. Experiment and result

4.1. Dataset and setting

Given the widespread popularity and public availability of the Hotel and Amazon datasets within the field of fake review detection, as well as their non-overlapping content (see Table 1), we have chosen these datasets for experiments. Table 2 indicates that each dataset already contains balanced real and fake reviews, so no rebalancing is required; and other statistics differ between the two classes.

Fig. 2 presents the distribution of review lengths, illustrating that the distributions of real and fake reviews are similar; and reviews in each dataset exhibit varying lengths, with the majority not exceeding 400 words. Comparatively, Amazon contains a higher proportion of shorter reviews than Hotel, but the differences at the sentence level are minimal, with the majority of reviews containing fewer than 50 sentences.

A concise analysis of the datasets' themes (see Fig. 3) shows that the content mirrors the nature of the its source platforms. For instance, in Hotel, both fake and real reviews frequently emphasize themes such as "room service" and "hotel," highlighting topics directly relevant to hospitality experiences. The reviews in Amazon often focus on topics like "quality" and "work great," emphasizing product performance and functionality. These underscore the diverse and platform-specific nature of the datasets, which is crucial for evaluating the generalizability of the proposed model.

In each dataset, 80% of the reviews are designated for training, and the remaining 20% for testing. We use F1-score (F1, %) as the main evaluation metric, followed by accuracy (Acc, %), and Area Under the receiver operating characteristic Curve (AUC, %). For the LLM, we employ the widely-popular LLAMA2-7B. We use the Parameter-Efficient Fine-Tuning, such as the Low-Rank Adaptation to train our LLMIC, with epoch = 5, batch size = 16, learning rate = 1e-5. The LLMIC model is loaded on two NVIDIA A100 GPUs (40GB), and consists of 6.6286e+09 parameters, of which 2.1277e+07 parameters are trainable, accounting for 0.321% of the total parameters. The average training time and inference time are 0.545 seconds and 0.544 seconds per Hotel review; and 0.657 seconds and 0.547 seconds per Amazon review. The reviews generation is performed offline and separately, so it adds no overhead to training time. In ablation studies, except for the module being replaced, the rest of the model architecture remains unaltered to maintain consistency in the experiment. Each experiment is replicated five times, with the average value as the result.

4.2. Effectiveness of LLMIC

To demonstrate the effectiveness of LLMIC, we conduct three distinct types of experiments:

- (1) To evaluate the overall performance of LLMIC.
- (2) To evaluate the effectiveness of XSMF (and MSFRP), we replace it within LLMIC with established or recent multifractal methods (and recurrence plots) to observe shifts in performance. Specifically, for multifractal methods, we test: (i) MFDFA (De-Santis et al., 2024; Stanisiz et al., 2024), which calculates the profile of the average embedding of a review and divides it into multiple scale-based segments. Each segment undergoes detrending and variance calculation, and the scaling behavior produces the multifractal sequence. (ii) HmF-DFA (Wang et al., 2023b), which optimizes MFDFA by introducing projection transformations and applying differential operations during detrending to ensure a stable fluctuation function. (iii) FS-MFA (Wang et al., 2024b), which performs Fourier series decomposition on the average embedding of a review, applies information entropy filtering to retain high-information components, constructs a historical volatility series from the profile sequence, and derives multifractal series through sliding-window detrend calculation and fluctuation analysis. For recurrence plots, there are: (i) Standard Recurrence Plot (SRP) (Stanisiz et al., 2024), which calculates pairwise distances between numerical points in the average embedding of a review. It applies a threshold to identify "close" points and marks them on a grid with both axes representing point indices, visualizing where the system's states recur. (ii) Cross Recurrence Plot (CRP) (Purohit & Panigrahi, 2024), which extends SRP by analyzing two random vectors from the review embeddings to identify cross-recurrences. (iii) Joint Recurrence Plot (JRP) (Stanisiz et al., 2024). which generalizes SRP by analyzing simultaneous recurrences among multiple random vectors in the review embeddings.

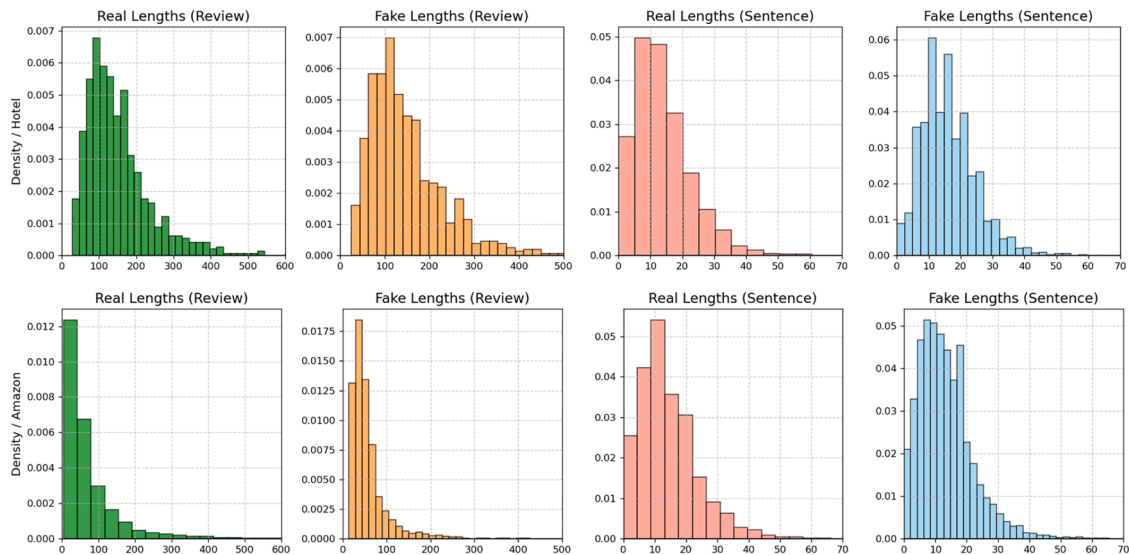


Fig. 2. Review length distribution, and sentence length distribution (word count).



Fig. 3. Phrase clouds across datasets, generated after applying data preprocessing steps, such as removing stop-words. Each phrase can be interpreted as a frequency-based topic, with larger phrases visually representing higher frequencies of occurrence within the dataset. To distinguish between different phrases, varying levels of color transparency have been employed.

- (3) We sequentially remove each implicit characteristic, Hurst space, Singularity space, Chi-space, RPoint, RLine, and RPlane, to gauge the individual impact on LLMIC performance. We also remove the Projector component and observe the resulting performance variations in LLMIC.
- (4) To examine the impact of different LLM backbones on LLMIC’s effectiveness, we replace the default LLAMA2-7B with two commonly-used alternatives: Gemma2-2B⁴ and Qwen2.5-14B⁵.

⁴ <https://huggingface.co/google/gemma-2-2b>

⁵ <https://huggingface.co/Qwen/Qwen2.5-14B>

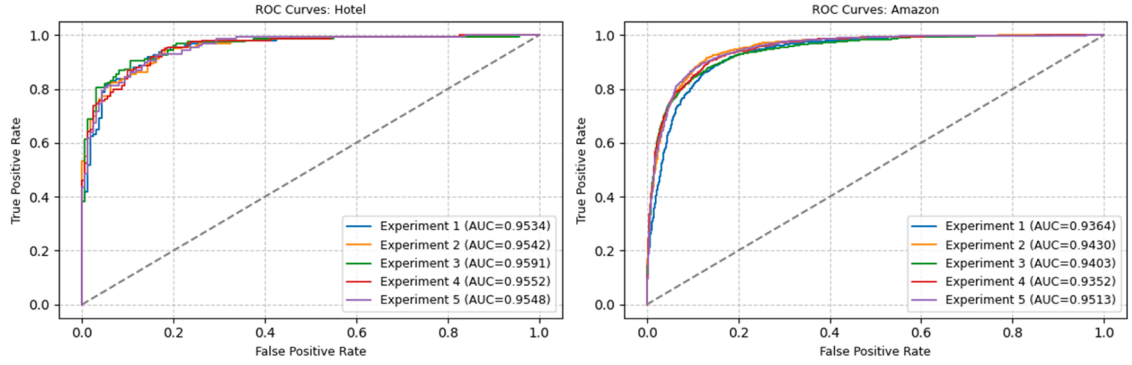


Fig. 4. ROC curves and confusion matrices for five experiments.

Table 3
Effectiveness evaluation results.

Model	F1	Acc	AUC
Hotel			
LLMIC (MFDFA)	94.27	94.08	94.16
LLMIC (HmF-DFA)	94.80	94.55	94.21
LLMIC (FS-MFA)	95.01	94.95	94.83
LLMIC (SRP)	94.85	94.06	94.07
LLMIC (CRP)	95.08	94.81	94.31
LLMIC (JRP)	95.00	94.16	94.39
LLMIC (without Hurst space)	93.97	94.14	94.45
LLMIC (without Singularity space)	94.74	94.31	93.91
LLMIC (without Chi space)	94.58	94.29	94.24
LLMIC (without RPoint)	94.15	94.61	94.36
LLMIC (without RLine)	94.21	94.41	94.47
LLMIC (without RPlane)	94.02	94.87	94.93
LLMIC (without Projector)	95.71	95.73	95.49
LLMIC (Gemma2-2B)	95.43	95.14	95.23
LLMIC (Qwen2.5-14B)	96.03	95.90	95.77
LLMIC (original)	95.89	95.88	95.53
Amazon			
LLMIC (MFDFA)	91.93	92.52	92.87
LLMIC (HmF-DFA)	93.54	93.29	93.02
LLMIC (FS-MFA)	93.57	92.74	93.11
LLMIC (SRP)	93.27	93.20	93.81
LLMIC (CRP)	93.46	92.90	93.70
LLMIC (JRP)	93.21	93.19	93.26
LLMIC (without Hurst space)	93.05	92.14	94.02
LLMIC (without Singularity space)	92.86	93.24	92.91
LLMIC (without Chi space)	92.83	92.19	94.11
LLMIC (without RPoint)	93.11	92.64	93.40
LLMIC (without RLine)	93.32	92.53	93.17
LLMIC (without RPlane)	92.99	92.89	93.22
LLMIC (without Projector)	93.70	93.69	94.02
LLMIC (Gemma2-2B)	93.59	93.22	94.04
LLMIC (Qwen2.5-14B)	93.74	93.81	94.36
LLMIC (original)	93.78	93.75	94.12

The ROC curve in Fig. 4 illustrates that LLMIC significantly outperforming the chance level (gray dashed line), which demonstrates its robustness in detecting fake reviews. Table 3 reports the remarkable capability of LLMIC; for instance, it consistently achieves over 93% in F1, accuracy, and AUC in identifying fake reviews on Amazon, with an impressive F1 of 95.89% on Hotel.

Replacing XSMF with MFDFA, HmF-DFA, or FS-MFA uniformly degrades LLMIC's performance on both datasets, underscoring XSMF's clear edge. Although MFDFA is widely utilized, it does not compete effectively in fake review detection. This is likely because it is more suited to traditional complex systems, whereas reviews are abstracted as complex systems. Moreover, MSFRP proves to be more effective than previous recurrence analysis methods, enhancing LLMIC's performance further. MSFRP's scale-based analysis is tuned to capture distinct characteristics, enabling the model to analyze reviews at varied granularities rather than at isolated points. This approach allows for the capture of different levels of recurrence behaviors, as illustrated in the diverse color spectrum of Fig. 10. Additionally, MSFRP factors in characteristics involving uncertainty within reviews, which is evident from the smooth color transitions, contrasting sharply with the black and white opposition typical of standard RP.

The results also show that removing any implicit characteristic results in a decrease in LLMIC's performance. For example, on Amazon, excluding the Hurst space results in a 0.7% decrease in accuracy. Moreover, the performance gains associated with each implicit characteristic are comparable, we believe that each surveys the authenticity of reviews from a unique perspective, revealing distinct strengths. The results also show that the removal of the Projector leads to a decline in LLMIC's performance.

When replacing LLAMA2-7B with Gemma2-2B, we observe a performance drop, likely due to Gemma2-2B's smaller parameter size, which limits its capacity for adequate review embedding and generation. Using Qwen2.5-14B results in only marginal performance improvement, suggesting that a larger model size does not necessarily guarantee significantly better outcomes. Considering the cost of deploying LLMs, models around the 7B parameter scale may offer a trade-off between efficiency and effectiveness for future applications.

It is worth emphasizing that the three evaluation criteria capture complementary aspects of performance. F1 harmonizes precision with recall, Accuracy measures overall correctness, and AUC reflects the quality of the predicted probabilities across every decision threshold. Across every experiment, the three metrics move in lockstep, rising and falling in unison—evidence that LLMIC's gains are comprehensive, not the artefact of tuning for any single score.

4.3. Competitiveness of LLMIC

We compare LLMIC with dozens of state-of-the-art models spanning every major detection paradigm (Table 1). Tables 4-5 show that LLMIC dominates across all metrics. On Amazon, LLMIC lifts F1 by 2.08%-19.43%, accuracy by 3.55%-17.28%, and AUC by 1.52%-17.33%; on Hotel, the gains are 2.53%-20.89% in F1, 0.38%-20.88% in accuracy, and 0.33%-13.65% in AUC. Notably, LLMIC surpasses the quantum neural network-based QuanNN by 9.78% F1 on Amazon; on Hotel, LLMIC exceeds DeceptiveBERT, a pre-trained language model tailored for detecting fake reviews, by 20.88% accuracy, and outperforms TransLSTM-RoBERTa by 2.73% accuracy. These results confirm that marrying large language models with complex-systems thinking offers a promising direction for fake review detection.

The success of LLMIC lies in its ability to leverage implicit characteristics, clues that often elude previous methods. Take three Amazon reviews that are superficially positive: FakeRoBERTa and the feature-driven IABSA-SS both misclassify them—Review #1 as genuine, Reviews #2 and #3 as fake—whereas LLMIC labels every one correctly, proving its sharper sensitivity to the fine-grained cues that matter.

Review #1: “My husband bought five for my daughter. They look so cute in actual and my baby looks very pretty when wearing them. Wonderful stuff for baby girls!” (fake)

Review #2: “Very nice quality. It is the modern way of tenderizing meat and so easy to use and clean. It comes with a cleaning brush also.” (real)

Review #3: “I love this kit, it's perfect for a beginner like me; I love the different colors and it's really good quality for such little value. Thank You!” (real)

4.4. Sensitiveness of XSMF

Scale δ is a critical parameter, as small and large scales distinctly influence the multifractal of reviews. In XSMF, δ is dynamically selected based on the power spectrum. To further explore its impact on XSMF, we observe fluctuations in LLMIC performance by varying δ .

For the analysis of δ_D (dimension direction; with D representing the number of dimensions), we fix δ_N (token direction; with N representing the number of tokens) = 2, maintaining the number of scales within each range consistent with XSMF at 20, starting

Table 4
Competitiveness evaluation results on Amazon.

Study	Model	F1	Acc	AUC
Salminen et al., 2022	OpenAI model	81.01	81.25	87.97
	FakeRoBERTa	83.21	83.06	87.11
	NBSVM	78.44	77.59	80.24
Hajek et al., 2023	LDA-ABSA	82.90	-	89.70
Liu et al., 2023	AugRoBERTa	84.15	82.94	-
	AugSVMTF	77.01	76.47	-
Birim et al., 2022	RFTD-VP	78.74	-	-
	DT-VP	74.76	-	-
	NN-VP	78.02	-	-
	SVM-VP	76.26	-	-
	GraphSAGE- Co	-	-	76.79
Narayan et al., 2022	DFNN	82.50	82.80	89.30
Hajek et al., 2020	DFSCNN	80.60	81.30	87.90
Zhang et al., 2024a	DynDistilBERT-BiLSTM-CNN	91.70	-	92.60
Bikku et al., 2024	QuanNN	84.00	86.00	-
Sharifpour & Lahmiri, 2023	ISIGRU	90.10	90.20	92.60
Rahim et al., 2024	IABSA-SS	74.35	77.24	77.04
Ours	LLMIC	93.78	93.75	94.12

Table 5
Competitiveness evaluation results on Hotel.

Study	Model	F1	Acc	AUC
Salminen et al., 2022	OpenAI model	86.75	86.25	93.74
	FakeRoBERTa	86.77	86.56	93.83
	NBSVM	84.02	83.13	92.58
Mohawesh et al., 2024	TransLSTM-RoBERTa	91.20	93.15	-
Hyder et al., 2024	DeceptiveBERT	75.00	75.00	-
Xu et al., 2022	WSUN	-	-	81.88
Lu et al., 2023	BSTC	93.36	93.44	-
Cao et al., 2022	ST-MFLC	88.00	88.00	95.20
Hajek et al., 2020	DFFNN	89.60	89.56	95.10
	DFSCNN	87.20	87.25	94.50
Liu et al., 2022	DSRHA	86.10	83.00	-
Han et al., 2023	EKI-SM	90.72	90.75	-
Mohawesh et al., 2023	EEG-SMC	92.08	93.60	-
Neisari et al., 2021	SOM-CNN	85.00	86.00	-
Bathla et al., 2022	CLSTM-ASP	87.00	95.50	-
Shunxiang et al., 2023	SIPUL	86.10	86.20	87.40
Singhal & Kashef, 2023	WSEM	92.00	88.00	-
Ours	LLMIC	95.89	95.88	95.53

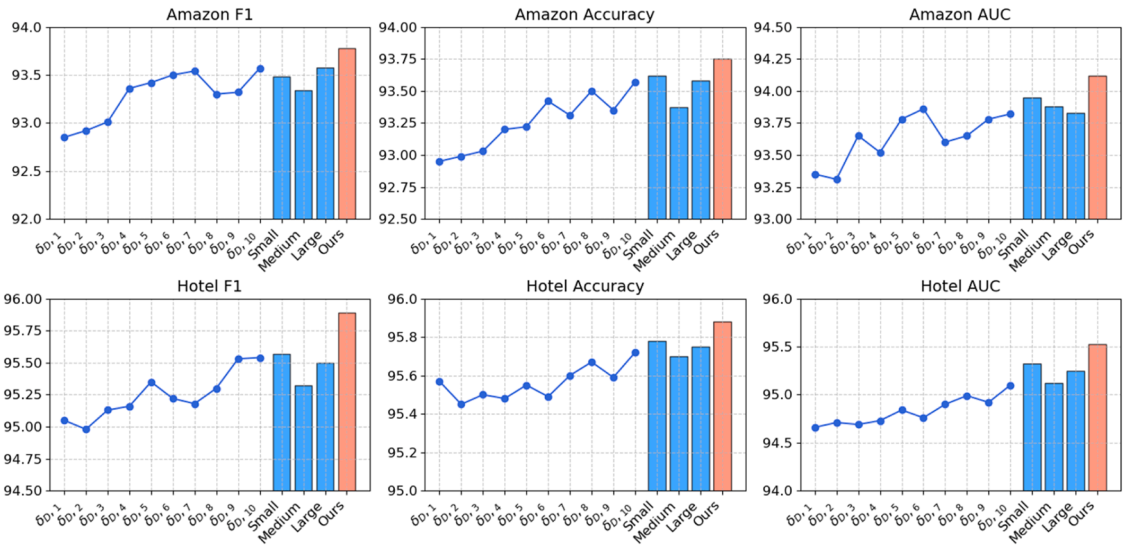


Fig. 5. Scale-based sensitivity analysis on Hotel.

initially with 5 dimensions and incrementally increasing. Specifically, we define the starting point s_i , and the endpoint e_i , for each list i ; the scale list $\delta_{D,i}$ includes all dimension scales from s_i to e_i . That is $s_i = 5 + 20(i - 1)$; $e_i = s_i + 19$; $\delta_{D,i} = \{s_i, s_i + 1, \dots, e_i\}$, where we select 10 lists for analysis, i.e., $i = [1, 10]$, achieving coverage from small (scale) to large (scale). For the analysis of δ_N , we fix δ_D at 10. Considering that the length (number of tokens) of reviews varies and is usually not long, we conduct experiments in tertiles: Small scale: $2 - (N/3)$ tokens; Medium scale: $(1 + N/3) - (N/3)$ tokens; Large scale: $(1 + N/3) - (N - 1)$ tokens.

Fig. 5 illustrates the superiority of our dynamic selection. Results also show that on the δ_D direction, larger scales tend to yield higher performance gains for the model. We conjecture that a larger δ_D means more dimensions of the embeddings are aggregated, reflecting more collective behavior and thus capturing a more comprehensive representation of the reviews' fractal. Additionally, it could average out noise and less informative variations across dimensions, as the model's trained fractal properties capture the characteristics of true and fake reviews, rather than overfitting to specific noises. Conversely, we find that smaller scales are often better on the δ_N direction. We believe that smaller δ_N , focusing on shorter text, enhances the model's sensitivity to local feature variations in detecting fake reviews. Moreover, it could aid in capturing short-term dependencies or local coherence between tokens.

4.5. Sensitiveness of MSFRP

The ε threshold, scaled by a multiplier η is a crucial parameter that determines the granularity of recurrence, and influences the quantification in MSFRP. We analyze the impact of η on LLMIC performance, typically varying η from 0.5 to 2. Additionally, to enhance our analysis, we include experiments with $\eta = 2.5$ and $\eta = 3$.

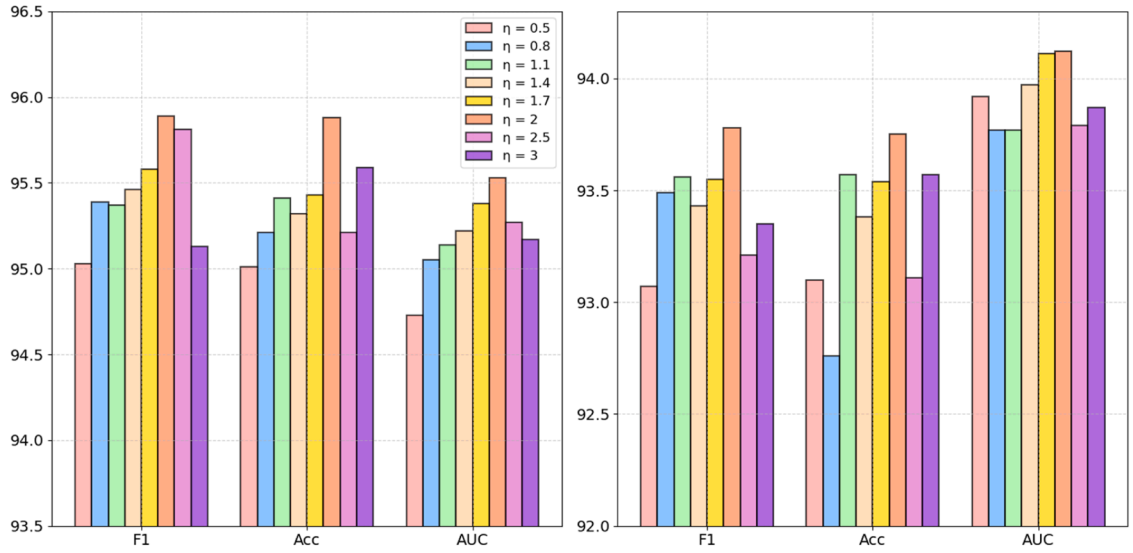


Fig. 6. Sensitivity analysis based on η , left-right: Hotel, Amazon.

Fig. 6 shows that as η increases, the performance of LLMIC generally shows an upward trend, peaking at $\eta = 2$. Increasing η beyond this point leads to a decline in LLMIC performance, suggesting a trade-off in the determination of the ε value. This behavior could be better understood as follows: a larger ε might reveal dynamics clearly but may also blur distinctions between closely spaced trajectories. Conversely, a smaller ε can sharpen these distinctions but risks losing weaker recurrences since the repetition would decrease, potentially underestimating the dynamics of chaos. On the other hand, considering noise sensitivity, a larger ε makes the recurrence analysis more robust to noise, as minor variations in state might still fall within the ε -neighborhood, but this leads to higher recurrence rates and more connected line structures, potentially exaggerating actual recurrent behaviors. Conversely, a smaller ε heightens the RP's sensitivity to noise, potentially contaminating the analysis with false recurrences or omit real ones due to slight noise-induced displacements.

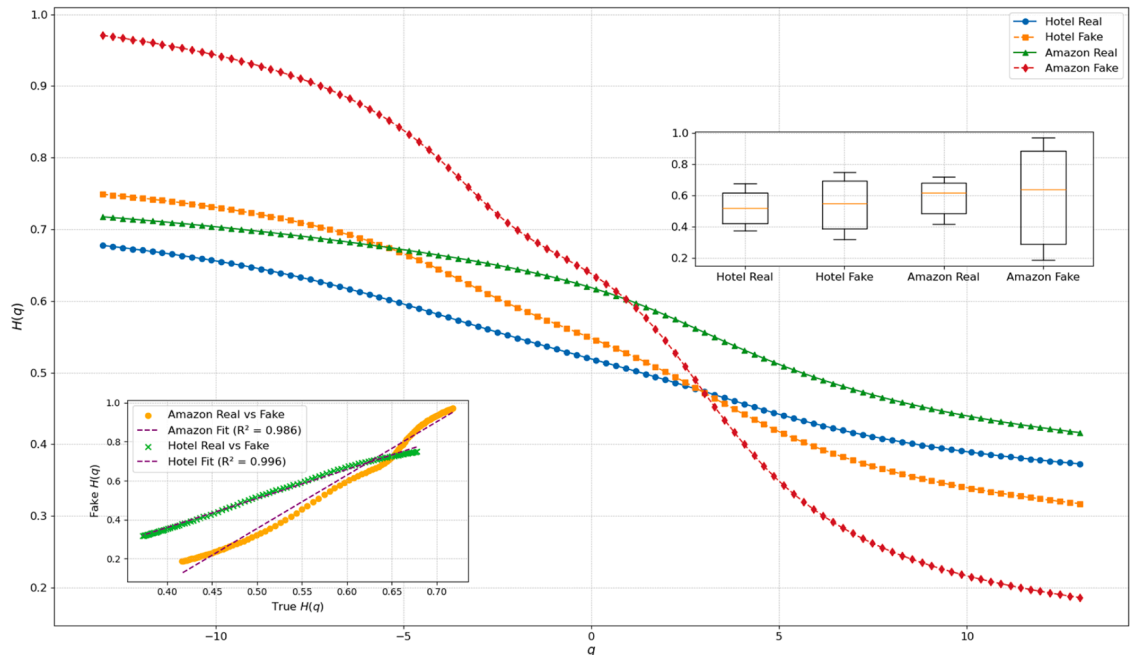


Fig. 7. XSMF's Hurst space. The box plot provides a summary of the distribution of Hurst exponents $H(q)$.

5. Discussion

This section further investigates the distinctions of real and fake reviews in terms of the implicit characteristics. For each dataset, we compute the average embedding of all real reviews and fake reviews respectively, and use these as representative inputs to generate the corresponding visualizations and provide a general discussion, such as highlighting the apparent differences between reviews.

From Fig. 7, we observe that each type of review possesses unique Hurst space characteristics. Firstly, the $H(q)$ values for real and fake reviews exhibit distinct patterns across different orders of q , with no overlap except at specific intersection points. As q increases, $H(q)$ for reviews decrease monotonically. The critical point $H(q) = 0.5$, marking the onset of self-correlation, is reached at different q values for each class.

Secondly, fake reviews exhibit higher maximum $H(q)$ values, indicating stronger long-term correlations and self-similarity in them. The minimum $H(q)$ values, representing stable states, are higher for real reviews, signaling stronger stability. According to the box plot, the breadth of the $H(q)$ distribution is broader for fake reviews across both datasets, implying that the inter-scale relationships in fake reviews are potentially more diverse.

Also, the subplot in the bottom-left indicates a positive correlation between the $H(q)$ values of real and fake reviews. This not only reveals the connection between real and fake reviews within Hurst space but also highlights the existence of the transformation itself as a distinguishing factor. These phenomena provide Hurst space-based insights for learning and differentiating the authenticity of reviews.

From Fig. 8, we observe that each type of review possesses its own characteristics within the Singularity space (i.e., no overlap virtually). Different reviews exhibit distinct $f(\alpha)$ and α values, with each point representing a unique multifractal structure. The Singularity space of different reviews forms irregular inverted U-shaped curves, each distinct in shape. This reflects the unique local complexities of the reviews, which may correspond to variations in linguistic patterns (Stanisz et al., 2024; Wang et al., 2024b). Across both datasets, the $\text{DoC}(\alpha)$ is consistently higher for fake reviews compared to real reviews, indicating a deeper complexity in fake reviews. While the $\text{DoC}(f(\alpha))$ does not show a uniform distinction between real and fake reviews, the differences in specific values reveal the richness of their respective distributions.

Fig. 8 also visualize $\Delta\alpha$ and $\Delta f(\alpha)$ on the right. On Amazon, $\Delta\alpha$ is consistently greater than zero, suggesting that the complexity of fake reviews deepens continuously compared to real reviews. On Hotel, this transition is slightly different; the deepening complexity shifts gradually from real to fake reviews. For $\Delta f(\alpha)$, it remains consistently below zero on Hotel, indicating that fake reviews exhibit a less complexity. On Amazon, this dynamic is slightly reversed, with the richness of complexity transitioning from fake to real reviews. These Singularity space-based findings hold promise for detecting fake reviews.

Fig. 9 demonstrates the characteristics of the two classes within the Chi-space. In the main plot each point corresponds to a pair of values (A_{LD} , D_G) for an individual review. The colored density regions represent the joint distribution of these two dimensions for different types of reviews. Darker shades indicate higher local density, while larger colored areas reflect broader dispersion in the space. It is observed that real reviews (in red) in Amazon display a broader range of D_G and A_{LD} values than fake reviews (in green),

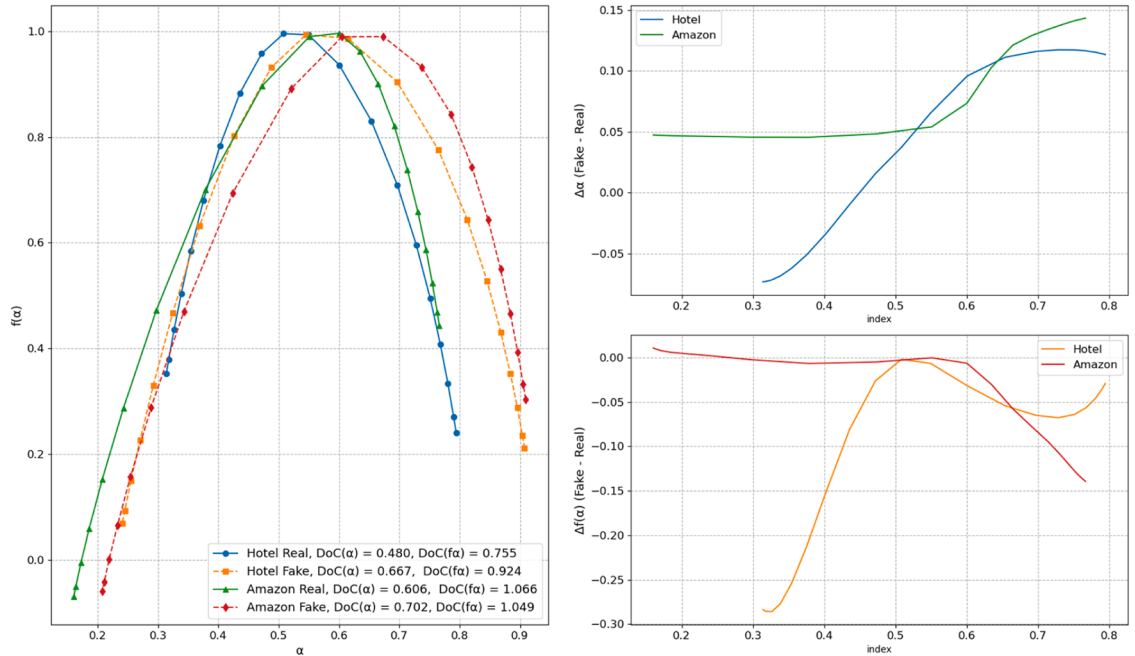


Fig. 8. XSMF's Singularity space. The degree of complexity (DoC) is quantified as: $\text{DoC}(\alpha) = \max(\alpha) - \min(\alpha)$, representing the depth of complexity, and $\text{DoC}(f(\alpha)) = \max(f(\alpha)) - \min(f(\alpha))$, representing the breadth of complexity.

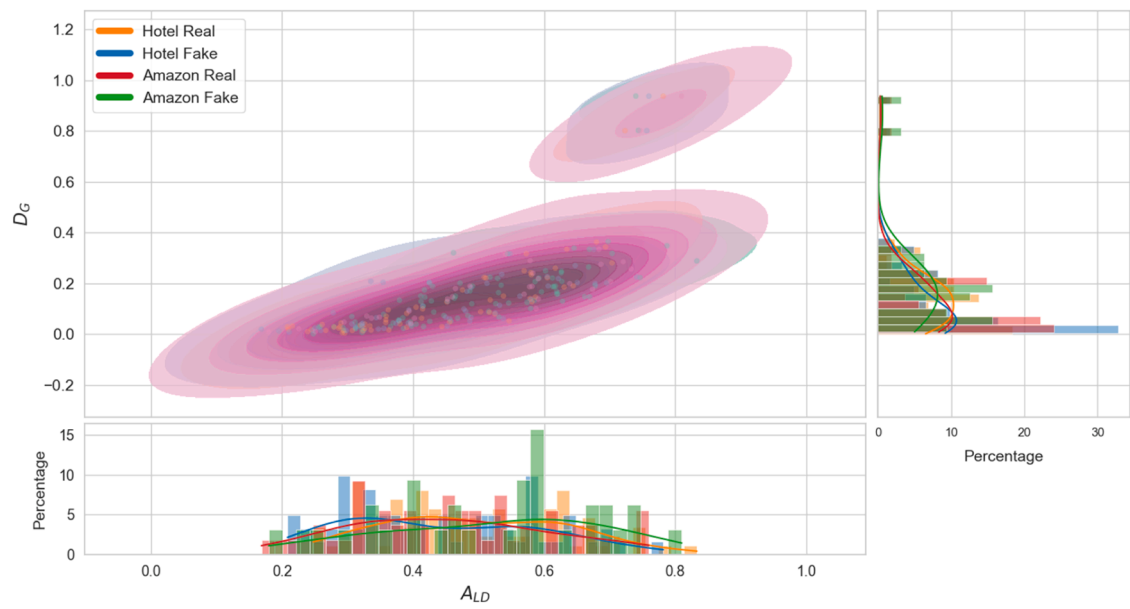


Fig. 9. XSMF's Chi-space.

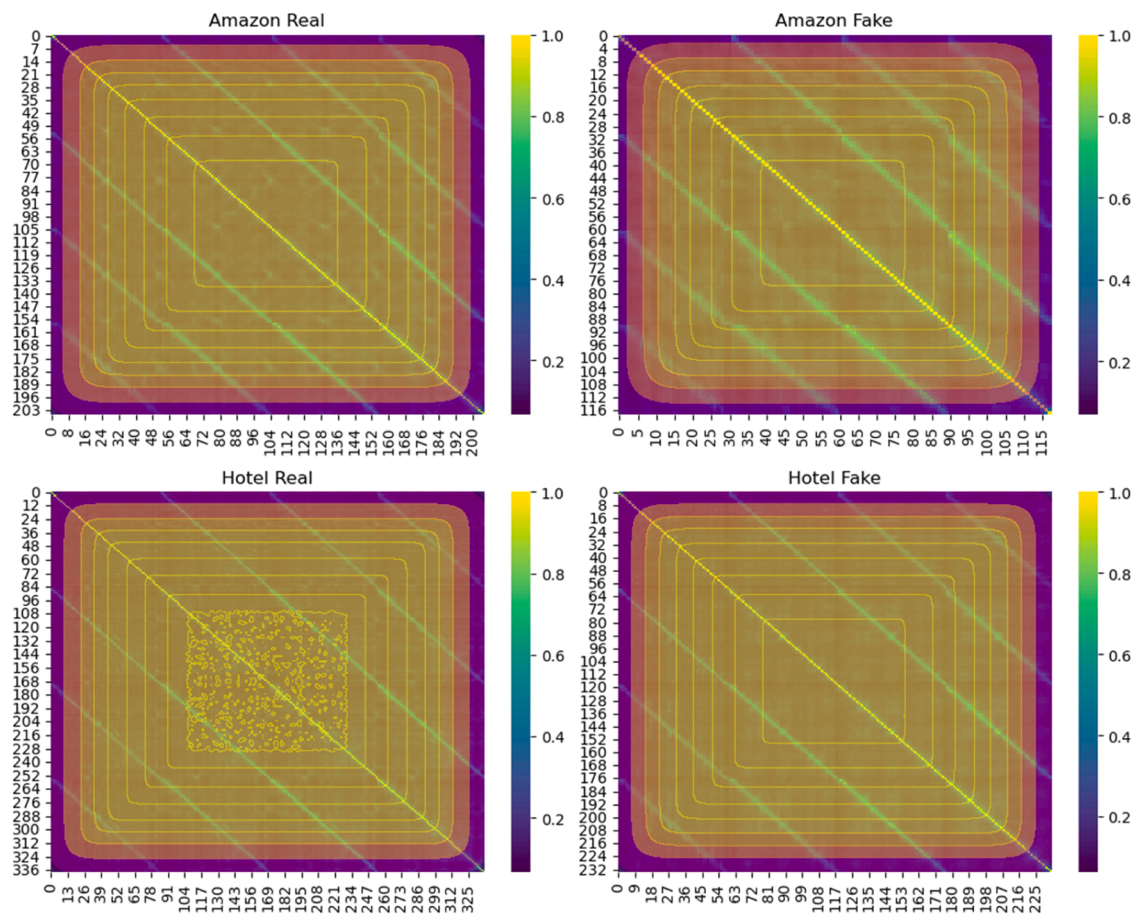


Fig. 10. Visualization of MSFRP.

which indicates that real reviews possess more diverse structural complexities within the Chi-space.

The kernel density estimation distribution of D_G values (the right subplot) reveals that fake reviews generally have higher density peaks than real reviews. For instance, on Hotel, the density peak for fake reviews exceeds 30%, whereas the peak for real reviews hovers around 20%. This suggests that fake reviews tend to cluster within specific regions of the Chi-space, reflecting potentially uniform structural patterns.

Fig. 10 shows MSFRP, a vibrant and diverse series of plots. It incorporates fuzzy computation to provide enriched feature information, which is evident in the smooth color transitions across the plots.

Recurrence points indicate recurrence behaviors in the token and embedding dimensions of the reviews. Although fake and real reviews exhibit similar recurrence values (as reflected in regions with similar colors), their coordinate positions for each recurrence value differ. Furthermore, the activity range of recurrence (the extent along both axes) is smaller for fake reviews, suggesting weaker relational connections between embedding values in fake reviews.

Recurrence lines reflect chaotic behaviors such as periodicity and cyclic patterns in the reviews. Fake reviews show more pronounced recurrence diagonals compared to real reviews. For example, the longest diagonal line in Amazon fake reviews is thicker, and in Hotel fake reviews, it is brighter. Additionally, the vertical and horizontal lines, which revisit the same or highly similar embedding values (often due to redundancy or a lack of variation), differ in length between real and fake reviews. For instance, within the smallest yellow rectangle on Amazon, the vertical span for embedding value index of real review is approximately 70-133, and the horizontal span is approximately 68-136. In contrast, for fake reviews, both spans are roughly 40-76.

For recurrence planes, both real and fake reviews show weaker recurrence intensity in the peripheral regions compared to the central areas, which likely reflects similarities in grammar and word usage. However, real reviews exhibit more irregularities than fake reviews. For instance, in the central region of Hotel reviews, real reviews display more irregular areas. Similarly, near the four corners of Amazon plots, real reviews show larger anomalous regions. This suggests that real reviews could contain greater variability in their content. These differences provide recurrence-based insights, which are potentially beneficial in fake review detection.

6. Implication

6.1. Theoretical implication

By positioning online reviews as complex adaptive systems, this study extends the theoretical dialogue between information science, AI and complexity theory.

First, we reconceptualize authenticity signals as implicit characteristics, i.e., macro-level, dynamic regularities that emerge from token interactions. Compared to prior work that examines cues such as styles, sentiment or embedding, our fractal and recurrence explorations reveal deeper insights, showing that fake and real reviews exhibit separable analysis in implicit characteristics. This moves the field beyond the long-standing assumption that deception often imitates visible language and instead suggests that fake reviews are constrained by hidden patterns (Liu et al., 2025).

Second, we integrate complexity science approaches into the LLM studies. Rather than merely fine-tuning encoders or hand-crafting features—as is common in spam detection, fake news detection, and e-commerce payment fraud detection (Li et al., 2025a; Li et al., 2025b; Zeng et al., 2025)—we regard the LLM's embedding manifold as the very arena where complex-system behavior unfolds. The XSMF and MSFRP demonstrate that high-dimensional contextual vectors preserve fractal geometry and recurrence structure. This finding implies that LLM embeddings carry both semantic and language dynamics, opening a new analytical avenue at the intersection of AI and nonlinear science.

Third, we couple LLMs with multifractal measures and recurrence plots in an end-to-end architecture, which expands the review search space (Sun et al., 2025), self-similarities, and cyclical/laminar states. This multi-layer coupling invites an examination of the match-mismatch idea of trust signals: authenticity may rest more on harmony between micro token and macro structure. Phenomena that unfold as a sequence of symbolic states may be amenable to the same multifractal and recurrence scrutiny. The core implication is that information processing systems, human or artificial, may benefit from attending to the fractal/recurrence shape of data streams, not merely to their literal content, when assessing veracity or risk.

6.2. Practical implication

This paper offers a promising extension to real-world tasks, such as AI-generated text (AIGT) detection (Wang et al., 2025b), cyberbullying detection (Cirillo et al., 2025c). Specifically, in AIGT detection, previous methods often focus on surface-level features such as word choice or syntactic structure, which can be easily mimicked by sophisticated AI. By applying XSMF, it becomes possible to identify the multifractal behaviors inherent in AIGT, revealing structures that differentiate AIGT from human-generated text (HGT). Furthermore, MSFRP captures recurrence patterns, recurring patterns characteristic of automated generation. This approach enables a more refined AIGT detection by uncovering patterns that go unnoticed by conventional methods. Similarly, in cyberbullying detection, challenges arise in detecting harmful interactions that often involve subtle shifts in tone or context, making them hard to differentiate from neutral communication. By applying XSMF and MSFRP, implicit characteristics of bullying behavior, such as recurring aggressive patterns or shifts in sentiment, can be identified. XSMF captures the fractal-like scaling of these behaviors, while MSFRP helps uncover recurrence patterns, revealing the dynamics of cyberbullying. This methodology not only improves detection accuracy but also contributes to a deeper understanding of how harmful online behavior unfolds, enabling better interventions and moderations.

In addition, this paper can be extended to other areas such as named entity recognition (NER) (Wang et al., 2022b; Wang et al.,

2025a); hazard event prediction (Zhang et al., 2024a) and sentiment analysis (Wang et al., 2024a). In NER, implicit characteristics could be used to model entity behaviors across different types, enhancing the extraction of subtle distinctions between entities. In hazard event prediction, XSMF and MSFRP can capture the multifractal nature and recurrence dynamics of risk processes, allowing for a more nuanced understanding of risk escalation. Similarly, in sentiment analysis, these methods could offer deeper insights into the shifting dynamics of sentiment in texts, uncovering the underlying structures of emotional discourse.

Furthermore, this paper offers implications for AI research. For instance, it encourages a rethinking of language representation by promoting architectures capable of explicitly handling the inherent complexity of language through fractal and recurrence-based representations (Muñoz et al., 2025). It also advocates for improved training strategies by incorporating constraint signals derived from implicit characteristics into contrastive learning frameworks, thereby enhancing model performance in complex, real-world tasks (Upadhyaya et al., 2025). Furthermore, it advances data augmentation by strategically prompting LLMs to generate high-quality, diverse, and realistic datasets tailored to various scenarios (e.g., fake news detection) and challenges (e.g., class imbalance) (Wang et al., 2023a; Ye et al., 2025; Sun et al., 2025).

7. Conclusion

This study introduces the idea of implicit characteristics to analyze online reviews and propose a novel method LLMIC to effectively detect fake reviews. Data experiments have confirmed the state-of-the-art performance of LLMIC, and the superiority of XSMF and MSFRP. We also contribute novel insights, including the operation of complex systems approach, and the differences of real and fake reviews in terms of implicit characteristics.

There are several limitations that future research could address. First, a more granular analysis is needed to explore how specific textual content, stylistic nuances, or vocabulary choices contribute to the overall complex system properties of reviews. For instance, understanding how constructs like metaphorical expressions, idiomatic usage, or context-specific terminologies influence the fractal and recurrence behaviors could provide deeper insights into the latent patterns distinguishing real and fake reviews.

Second, generating effective and diverse training reviews remains a critical area for improvement. One promising direction is leveraging Generative Adversarial Networks to synthesize fake reviews that are particularly challenging for detection models. Another approach could involve introducing adversarial perturbations into generated reviews to simulate edge cases or outliers, enhancing the model's resilience to deceptive tactics. Active learning could also be employed to iteratively refine generated labels, enabling the model to focus on learning from reviews with the most informative or uncertain characteristics. Additionally, assigning differentiated importance weights to generated reviews based on their informativeness or alignment with real-world distributions could enhance training effectiveness.

Third, while the universality of XSMF and MSFRP in capturing the complex and dynamic attributes of text suggests strong adaptability to diverse datasets, this generality also poses challenges. Content variation across domains introduces new intricacies, such as domain-specific linguistic structures, cultural context, or ethical constraints, which could impact the model's generalizability. Future research should evaluate LLMIC on datasets from diverse fields, including medical reviews, legal opinions, and academic feedback. Further, there is a need to explore whether these implicit characteristics could serve as universal markers for text authenticity or even extend to credibility of information, such as evaluating misinformation, may expand the boundaries of understanding information.

Furthermore, automated fake-review detection carries ethical challenges. For example, misclassifying a genuine review may suppress legitimate speech and unfairly harm honest merchants; subtle demographic or linguistic bias may quietly exclude entire user groups; indiscriminate text inspection erodes privacy; and a powerful detector could be weaponized for censorship. Mitigation must pair technical safeguards with clear governance. Specifically, we recommend: (1) confidence-based triage, with low-confidence outputs acted on automatically, and borderline cases flagged for human review; (2) a transparent appeal mechanism that lets users contest decisions; (3) periodic bias and performance audits across demographic lines, etc.

CRedit authorship contribution statement

Zhenhua Wang: Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Aixin Yao:** Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization. **Guang Xu:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Ming Ren:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition.

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Data availability

Data will be made available on request.

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